TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
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2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***Development of ECHO PROTOCOL: An Online Cooperative Horror Game with Adaptive AI and a Client-Server Virtual Item Shop***

3.1.2. Vietnamese: ***ECHO PROTOCOL: Trò chơi kinh dị hợp tác trực tuyến tích hợp AI và hệ thống cửa hàng vật phẩm ảo theo mô hình Client-Server***

(*) 3.2. Main proposal content (including result and product)

a) Context:

The game industry provides a practical environment for applying software engineering knowledge in game development, artificial intelligence, real-time networking, database systems, user interface design, and 3D content creation. Online cooperative horror games require players to communicate, share responsibilities, respond to unpredictable threats, and make decisions under pressure.

ECHO PROTOCOL is a first-person online cooperative survival-horror game for one to four players on Windows computers. The registered playable scope is one Expedition mode in one abandoned research facility. Players must coordinate to explore a hostile environment, complete interdependent objectives, survive a pursuing enemy, and escape.

In Expedition, players locate energy cores, restore power and security systems, complete cooperative environmental interactions, and unlock the evacuation sequence. A successful match requires completion of the defined objectives and a valid escape; failure and player elimination follow the mission rules. This single, fully integrated gameplay loop is the primary focus of the project.

The horror experience is built around one primary Stalker enemy with patrol, perception, investigation, pursuit, search, and attack behaviours. Player noise and visible actions provide detection cues. A bounded Adaptive AI mechanism uses observed gameplay information to adjust selected parameters within safe limits, with a fixed configuration when adaptive input is unavailable. Its effect will be evaluated rather than presumed.

The Unity and C# Windows client uses Photon Fusion for cooperative multiplayer. An ASP.NET Core Web API deployed on AWS manages accounts, player progression, match results, ECHO Credits, and a client-server virtual item shop. ECHO Credits are earned through validated gameplay rewards or seeded for controlled tests. Real-money payments, a separate player web portal, additional game modes or locations, and built-in voice chat are outside the mandatory registered scope.

b) Proposed Solutions:

- Develop and complete one first-person Expedition experience in the abandoned research facility for one to four online players, with private rooms and selectable supported difficulty levels.
- Integrate the Expedition objective chain: locate energy cores, restore designated systems, complete cooperative tasks, and activate evacuation only after required objectives are met.
- Synchronize players, interactions, objective progress, enemy states, player condition, and win/loss outcomes under host authority; test valid and interrupted multiplayer sessions.
- Complete perception-based behaviour for one primary Stalker enemy and integrate at least one bounded Adaptive AI adjustment with a verifiable fixed-baseline fallback and recorded decision evidence.
- Complete account authentication, match rewards, the ECHO Credits wallet, an in-game shop, inventory, and approved cosmetic or supporting items, with server-side purchase validation.
- Deploy and integrate the backend and a limited web administration interface on AWS. Protect persistent game and economy data. No real-money purchase, payment-provider checkout, or player-facing payment portal is required for the registered product.

c) Functional Requirements:

Multiplayer Session and Lobby System

- The game shall host private Expedition sessions for one to four players. A host may select a supported difficulty; a client may join using valid room information. Invalid or full-room join requests shall be rejected without changing the current session.
- The lobby shall show player identity and ready status. Only the host may start a match, and every connected participant shall load the same Expedition mission. Non-host start requests shall have no effect.
- During a match, the host shall authoritatively synchronize player life states, objective completion, enemy activity, and the final result. On connection loss, the affected client shall receive a reconnect or session-ended status; advanced host migration is not required.

Player and Cooperative Gameplay System

- Players shall walk, sprint, crouch, interact with mission objects, carry inventory items, use UV and other defined equipment, and send available in-game assistance signals. Actions shall respect inventory and interaction eligibility rules.
- Movement or interactions that emit noise shall be available to the Stalker perception system. A Downed player may be revived only by an eligible teammate with the required medical resource and within configured limits; an invalid revive attempt shall not restore the player.
- Expedition shall require energy-core collection, power and security objectives, and cooperative interactions before escape is enabled. The host shall reject premature evacuation and commit a single match outcome when the final condition is met.

Expedition Mission and Level Content

- The deliverable shall contain one playable abandoned research facility environment with a complete Expedition objective sequence, reachable interaction points, enemy navigation, and a defined evacuation location.
- Objective progress shall follow the configured mission sequence. Completed objectives shall be reflected consistently on all connected clients; the escape interaction shall remain unavailable until its prerequisites are satisfied.
- The mission shall support Win and Lose outcomes, including failure when the team cannot complete the escape under the defined match conditions. Additional environments and a Purge mode are excluded from the mandatory deliverable.

Monster AI and Adaptive AI System

- One primary Stalker enemy shall navigate the facility using patrol, investigation, pursuit, search, attack, and recovery behaviours. Gameplay tests shall verify state changes after valid stimuli and that the enemy can return to patrol or search after losing a target.
- Enemy detection shall be based on observable vision, noise, and recent encounters rather than unrestricted player-position knowledge. UV equipment shall interrupt the enemy only when configured range, resource, and cooldown conditions are met.
- Adaptive AI shall use supported observed player or team metrics to apply at least one bounded adjustment without exceeding a configured minimum or maximum. Missing, invalid, or insufficient input shall produce a fixed-baseline fallback instead of an unvalidated gameplay change.
- For comparable Fixed and Adaptive runs, the system shall record match identity, selected mode, applied or fallback decision, and relevant outcome metrics. The team shall provide test cases and comparison results without claiming effectiveness before evaluation.

Account and Authentication System

- Players shall be able to register, log in, log out, and retrieve their own profile. Invalid credentials shall be rejected and unauthorized requests shall not access another player's data.
- Authenticated clients shall retrieve player progression, ECHO Credits balance, inventory ownership, and selected customization from the backend. Values shown in the client shall match accepted server records.
- The backend shall hash passwords, validate access tokens, and authorize critical operations. Client-supplied balances, rewards, item ownership, and match outcomes shall not be accepted without the required server checks.

Virtual Shop Frontend

- The in-game shop shall display server-provided ECHO Credits balances, catalogue entries, item prices, descriptions, and owned status for the supported item catalogue.
- Players shall browse the catalogue, purchase an eligible item using ECHO Credits, view acquired items in inventory, and equip compatible items. Unowned or incompatible items shall not be equipped.

- The mandatory catalogue shall support defined customization items and non-pay-to-win supporting equipment. Companion pets, extensive cosmetic collections, and other optional content are not required for acceptance.

Shop Backend, Wallet, Inventory, and Administration

- Authenticated backend services shall manage profiles, wallets, shop products, purchases, inventories, and equipped items. The limited administration interface shall allow authorized users to view user, wallet, and purchase records and perform auditable permitted wallet adjustments.
- Every purchase shall check product availability, ownership eligibility, and sufficient ECHO Credits on the server. A successful purchase shall deduct credits and grant ownership atomically; rejected or repeated requests shall not produce inconsistent or duplicate grants.
- ECHO Credits shall be granted for validated Expedition match results according to a configured reward rule. Retrying the same completed match result shall not issue the reward a second time.
- Players shall purchase shop items only with ECHO Credits. Credits may originate from validated gameplay rewards or explicitly authorized test/admin allocations; no cash purchase or external top-up is part of the required workflow.

User Interface, Results, and Data Logging System

- The Windows client shall provide authentication, a main menu, private-room lobby, mission display, settings, in-game HUD, inventory, item shop, and result screen. A separate Player Web Portal is outside the mandatory scope.
- The HUD shall indicate current Expedition objectives, interaction prompts, teammate condition, mission item status, UV/resource availability, and evacuation information when relevant.
- At match completion, the result screen shall show Win/Lose, mission progress, survival information, and accepted ECHO Credits or XP rewards. A committed match shall produce one authoritative result record.
- Backend persistence shall cover accounts, progression, wallets, shop purchases, inventory, match outcomes, and AI evaluation data. Access to sensitive records shall be authenticated and role-appropriate.

Economy Boundary and Transaction Integrity

- The registered economy shall use virtual ECHO Credits earned through gameplay or provisioned for controlled testing. No real-money payment gateway, payOS top-up, or commercial payment flow is required. Shop deductions, reward grants, and administrative wallet changes shall be server-validated and auditable.

d) Non-Functional Requirements:

- The standalone game client shall run on Windows 10 or Windows 11 (64-bit) with a selectable 1920 x 1080 Low graphics preset.
- The performance target is an average of at least 30 FPS at 1920 x 1080, Low preset, during a representative 15-minute Expedition multiplayer test with up to four players. Record frame-time/FPS results from the Windows build on the minimum target hardware; the target is provisional until benchmarked.
- Proposed PC configurations (subject to play-test verification): minimum target - Intel Core i5-8400 or AMD Ryzen 5 2600, 16 GB RAM, NVIDIA GTX 1060 6 GB or AMD RX 580 8 GB, and 15 GB free SSD space; recommended test configuration - Intel Core i5-10400F or AMD Ryzen 5 3600, 16 GB RAM, NVIDIA GTX 1660 SUPER 6 GB or AMD RX 5600 XT 6 GB, and SSD storage. A stable broadband connection is required for online play.
- Photon Fusion shall manage host-authoritative sessions for one to four players. Host validation shall cover objective completion, player rescue, AI damage, and match results; test clients shall agree on the committed mission state.

- Backend and limited administration services shall be deployed on AWS with role-controlled access. Persistent records and credentials shall be protected; backup and recovery procedures shall be verified in the test environment.
- Real-time multiplayer shall be separate from persistent backend APIs. The host shall coordinate gameplay while the backend manages verified accounts, profiles, match records, ECHO Credits, and owned items.
- Authentication shall use hashed passwords, protected tokens, role-based authorization, and server-side checks. Requests without sufficient permissions shall be rejected by the applicable API endpoints.
- The virtual shop shall verify prices and credits and record wallet deductions and item ownership consistently. Duplicate requests and insufficient-balance requests shall not create duplicate ownership or negative wallet balances; no real-money processing is included.
- The software shall remain modular and testable. Third-party game assets shall follow applicable licenses, and performance, multiplayer, AI, and transaction tests shall be documented with reproducible conditions.

e) Theory & Practical:
- Software Development Life Cycle (SDLC), iterative Agile development, requirements management, and software quality assurance.
- Object-oriented programming and component-based game development using C# and Unity.
- Online multiplayer architecture, state synchronization, network authority, and real-time communication.
- Finite State Machines, behaviour-based enemy decision-making, gameplay telemetry, bounded Adaptive AI, fixed-baseline fallback, and comparative testing methods.
- Field-of-view and line-of-sight detection, noise-based perception, NavMesh pathfinding, and spatial decision-making for enemy navigation.
- RESTful API design, client-server integration, HTTP communication, JSON data exchange, and error handling.
- AWS deployment, RESTful service integration, secure data storage, and gameplay event processing without real-money payment processing.
- Authentication, authorization, password hashing, access-token validation, and secure management of player data.
- Game Design Document preparation, cooperative puzzle and level design for one facility, horror pacing, interaction design, and player experience principles.
- Unity gameplay and AI tests, multiplayer integration tests, backend and economy tests, Fixed-versus-Adaptive evaluation, and repeatable Windows performance measurement.

f) Products (Expected Deliverables):
- One playable Windows first-person cooperative horror game supporting one to four players in a complete Expedition mission.
- A host-controlled private-room lobby with room creation, joining, readiness, supported difficulty choice, and synchronized mission start.
- One synchronized cooperative survival system for movement, mission interactions, player life states, item use, enemy pressure, and results.
- One playable abandoned research facility with energy-core, power/security, cooperative objective, and evacuation interactions, including defined Win/Lose outcomes.
- One primary Stalker AI with perception, navigation, investigation, chase, search, attack, recovery, and testable counterplay using equipment.
- A bounded Adaptive AI feature using observed gameplay records, a fixed-baseline fallback, and recorded decisions for comparative evaluation.

- An ASP.NET Core backend deployed on AWS with authentication, player progression, validated match rewards, ECHO Credits wallets, inventories, and persistent records.
- An in-game virtual item shop and a limited web administration interface for authorized account, wallet, item ownership, and purchase management.
- Match and AI evidence, a reproducible Fixed-versus-Adaptive evaluation plan and results, and documented economy integrity and multiplayer integration tests. Additional modes, maps, voice chat, Player Web Portal, and payOS are not mandatory deliverables.
- Project deliverables comprising the GDD, SRS, software architecture and diagrams, source code, test cases and results, deployment instructions, user guide, demonstration video, and Windows build with documented hardware and FPS measurements.

g) Proposed Tasks:

- Task package 1: Finalize the GDD, SRS, architecture, one-map scope, and testable feature/NFR acceptance criteria.
- Task package 2: Complete multiplayer lobby and host-authoritative mission integration; test joining, readiness, disconnects, and outcomes.
- Task package 3: Finish player movement, equipment, inventory, Downed/Revive rules, and replicated cooperative interactions.
- Task package 4: Complete and balance the Research Facility Expedition mission, including objectives, puzzles, evacuation, and Win/Lose handling.
- Task package 5: Integrate and refine Stalker perception, navigation, pursuit/search states, UV counterplay, and related diagnostics.
- Task package 6: Complete bounded Adaptive AI integration, profile inputs, Fixed Fallback, telemetry, and comparative gameplay tests.
- Task package 7: Integrate and secure backend accounts, player progression, mission rewards, and ECHO Credits persistence on AWS.
- Task package 8: Complete the in-game shop and limited administration flow, including purchase validation, inventory ownership, and wallet audit records.
- Task package 9: Implement reproducible Unity, multiplayer, API, economy, and AI regression tests; measure 30+ FPS on reference hardware.
- Task package 10: Optimize gameplay and presentation, package the Windows build, and deliver documentation, deployment instructions, and demo evidence.

4. Other comments (propose all relative things if have):

The proposed Review 2 scope is one Expedition mode, one Research Facility environment, one primary Stalker enemy, one-to-four-player co-op, bounded Adaptive AI, and a client-server ECHO Credits item shop.

Additional modes and maps, supporting enemy variants, built-in voice chat, a separate Player Web Portal, and payOS or other cash top-ups may remain in source as exploratory work but are excluded from registered acceptance and delivery commitments.

A provisional 30+ FPS target and reference PC specifications are documented above; actual hardware suitability must be verified on a Windows build. The remaining scope will undergo integration, security, gameplay, AI, and multiplayer testing before final evaluation.

HCM, date 21/10/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*